

TRI

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□ □ □ □ □ Updating the World Is Not Rebinding the Target: Temporal Referent Integrity in Tool-Using Agents

□ □ □ □ □ AAIL 2027 Main Technical Track

□ □ □ □ □ 2026 7 21

□ □

“ ” “ ”

Temporal Referent Integrity TRI

SQLite

TRI

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API

ID

A Agent

B

“

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“

”

“ ”

A

ID

“

”

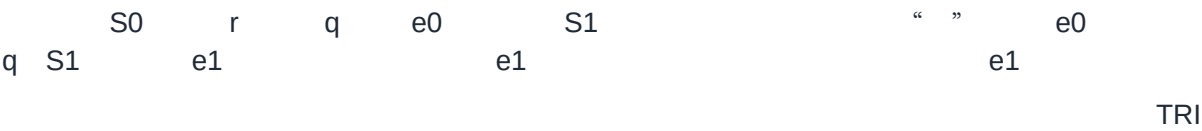
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- 1. “ ” “ ”
- 2.
- 3.



2.1



2.2



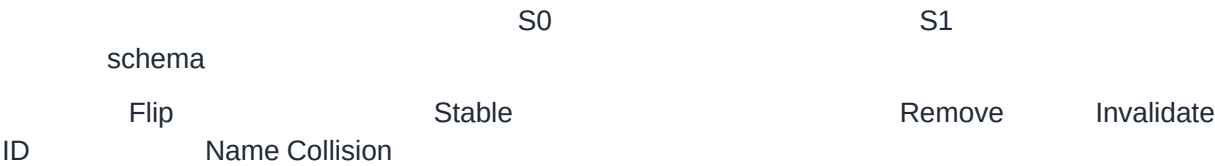
2.3



3.1

- 4.
- 5. TRI
- 6. gate
- 7. schema
- 8.
- 9. Agent benchmark

3.2



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160 20 8 240 40 10 schema Qwen GLM
DeepSeek

3.3

actor gate ID
reference mode gate Agent
CTA
Binding Drift AWS Bedrock re-verification
author adaptation Binding Drift

3.4

TRI Stable TRI
10,000 bootstrap McNemar API intention-to-treat
thinking token

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4.1

160 Qwen GLM 64.4% 71.9% 95.0% 96.2%
gate 98.1% 100.0% ID

	Qwen3.5	GLM-5.1
	64.4%	71.9%
	95.0%	96.2%
actor	96.9%	98.1%
+ gate	98.1%	100.0%

60.0% “ ” “ ”

4.2

/ 75.0% 81.2% 70.6% gate 65.0% 73.1%
CTA actor 96.9% 98.1% gate 1.2 1.9

	Qwen3.5	GLM-5.1
+ reference mode	75.0%	75.0%

TRI

	Qwen3.5	GLM-5.1
	81.2%	70.6%
+ validity gate	65.0%	73.1%
	95.0%	96.2%
actor	96.9%	98.1%
+ gate	98.1%	100.0%

gate

4.3 240

59/79	schema CTA	0	Qwen	GLM	DeepSeek	43/72	38/80
			CTA	CTA			
Qwen3.5		43/72	0/71	+23.3			
GLM-5.1		38/80	0/70	+24.2			
DeepSeek		59/79	0/70	+17.5			

10,000 0 CTA 10 100% 20 100% 10

4.4

ledger	240	Agent	TRI
			CTA
Qwen3.5	63.3%	69.6%	70.8%
GLM-5.1	67.1%	80.8%	94.2%
DeepSeek	68.8%	75.8%	91.2%

DeepSeek Qwen 1.2 80 0 56 38 42 CTA GLM 0

4.5 SQLite

ID		40	SQLite		Qwen	GLM	13	8
gate	0	140						
1,440		episode	Qwen	GLM	DeepSeek	87	79	75
70	46	57	“		” validity gate			

4.6

50		100		
gold	Fleiss kappa	0.708	benchmark gold	86%
50	Qwen/GLM		60%/74% CTA	90%/98%
Preserve/Reevaluate			fallback	

4.7 schema

80	schema			Qwen	46.2%	82.5%
schema	selector grounding	40		guard		guard
				Qwen	35/40	39/40 GLM

4.8 Binding Drift

Binding Drift			Entity Lock			TRI		
20	Entity Lock		10/10	0/10	GLM self-reverify	0/10	10/10	
			Qwen reverify	3/20	grounding	CTA	Qwen	GLM
12/20	17/20							
prompt		TRI	verifier	“	”	Binding		
Drift	CTA	Binding Drift						

4.9 benchmark

	ToolSandbox	AppWorld	tau3-bench	ToolSandbox	129	
AppWorld	732	244	near-match	tau3-bench	2,449	10,832
AppWorld		28	TRI			
	TRI			benchmark	“	
	”	ToolSandbox			GLM	3/6
Stable	0/2					

TRI



benchmark

benchmark AAAI artifact

CTA SQLite



6.1

Agent TRI TRI “ ” LLM
ToolSandbox “ ” “ benchmark ”

6.2

“ ” “ ” TRI Binding Drift
Entity Binding Failures Agent

6.3

Binding Drift

6.4

160 240

6.5

“ benchmark ” TRI strict/near-match/excluded

6.6

CTA selector capability

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“ ” “ ”

Binding Drift

AAAI 2027



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